

Electromagnetic Radiation/Electron

Wave or particle ??

Particles

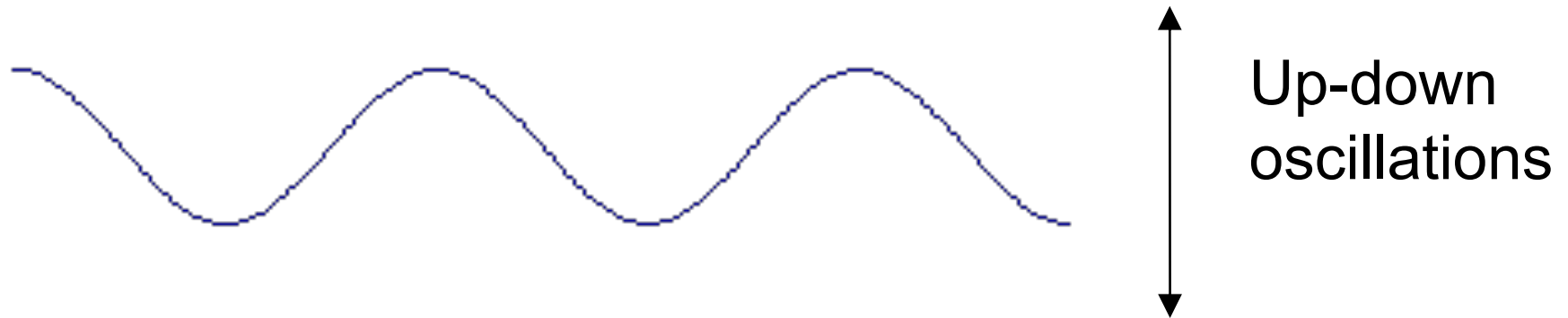
Base Ball, Car, a grain of sand are examples of **particles**.
They can be located at a well-defined point at a given time.
They can be at rest, moving or accelerating.

Falling Ball



Waves

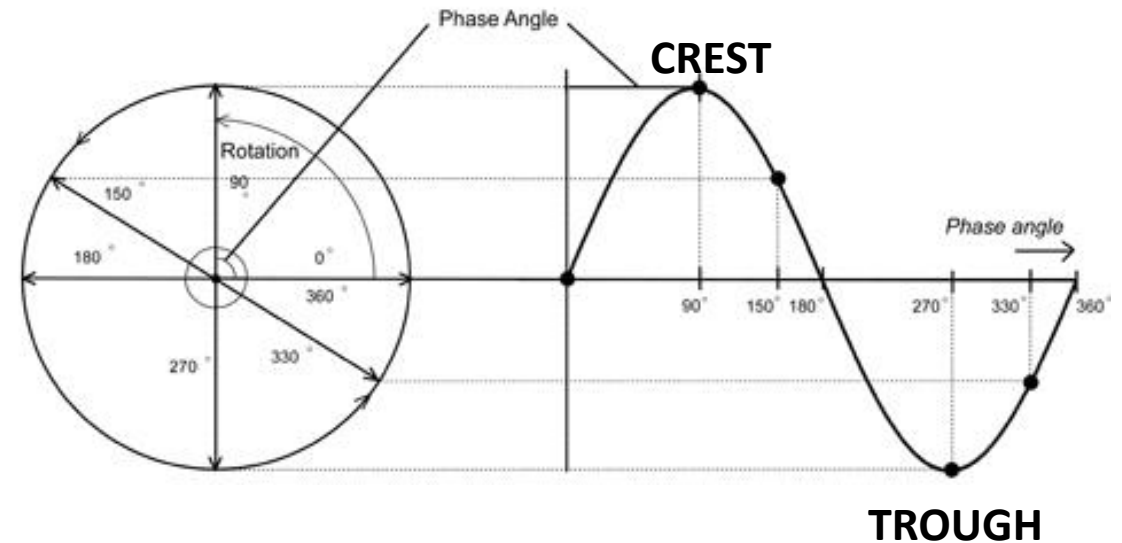
Waves are oscillations in space and time.
They are delocalized.



A wave travelling in x-direction

Wave

- A wave repeats regularly in space.
- A wave may be regarded as the ray trace of a combination of uniform forward motion and Uniform circular motion.

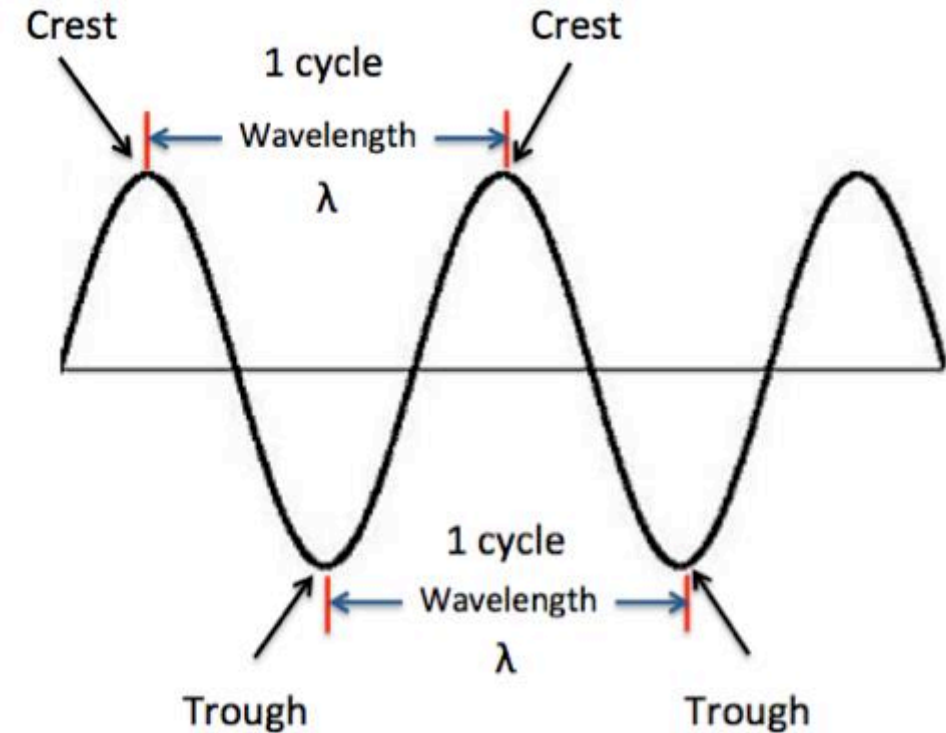
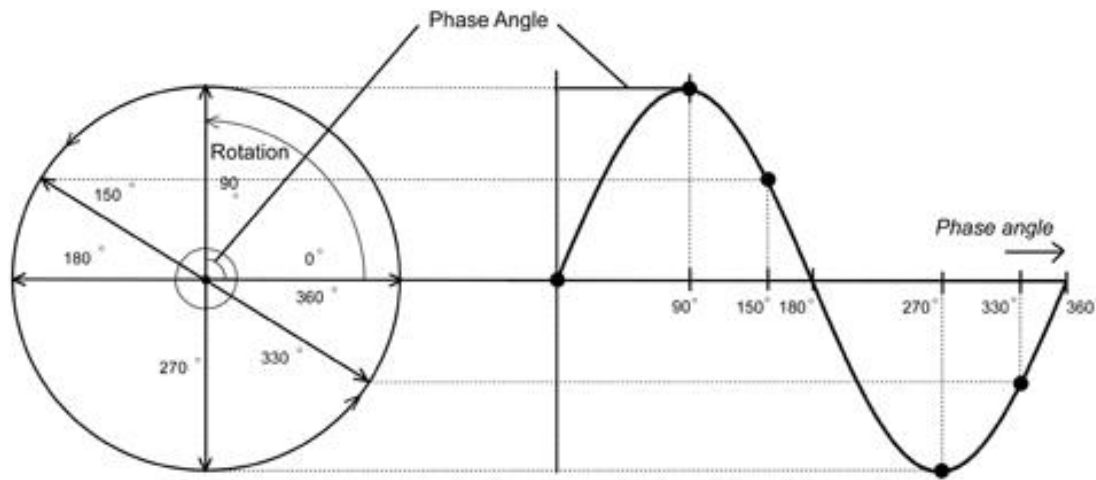


Change in Horizontal displacement with time is similar to Uniform forward motion

Change in vertical displacement with time is similar to Uniform circular motion.

<https://physics.stackexchange.com/questions/142888/wave-displacement-with-phase-difference-180>

One cycle



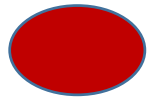
Crest, Trough and Wavelength

<https://physics.stackexchange.com/questions/142888/wave-displacement-with-phase-difference-180>

<https://study.com/academy/lesson/wave-period-definition-formula-quiz.html>

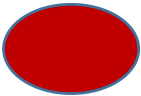
Particle

- When a particular space is occupied by one particle, the same space cannot be occupied simultaneously by any other particle. In other words, particles do not interfere.
- A particle can displace other particle and occupy the particular location in space previously occupied by the other particle.



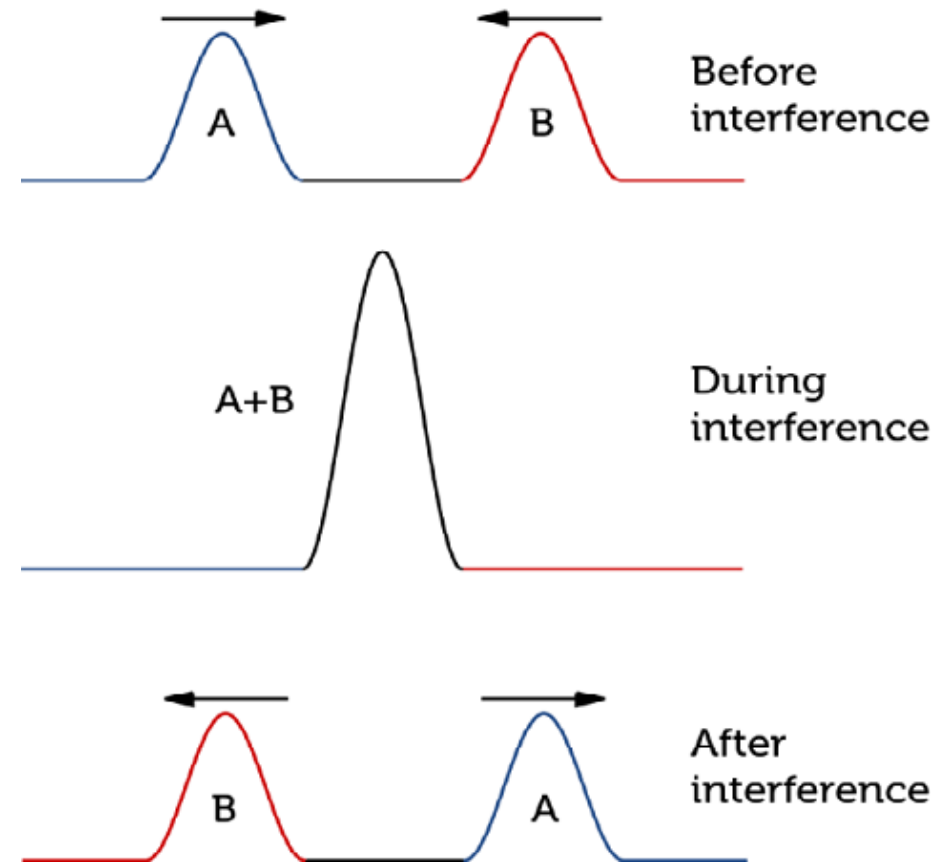
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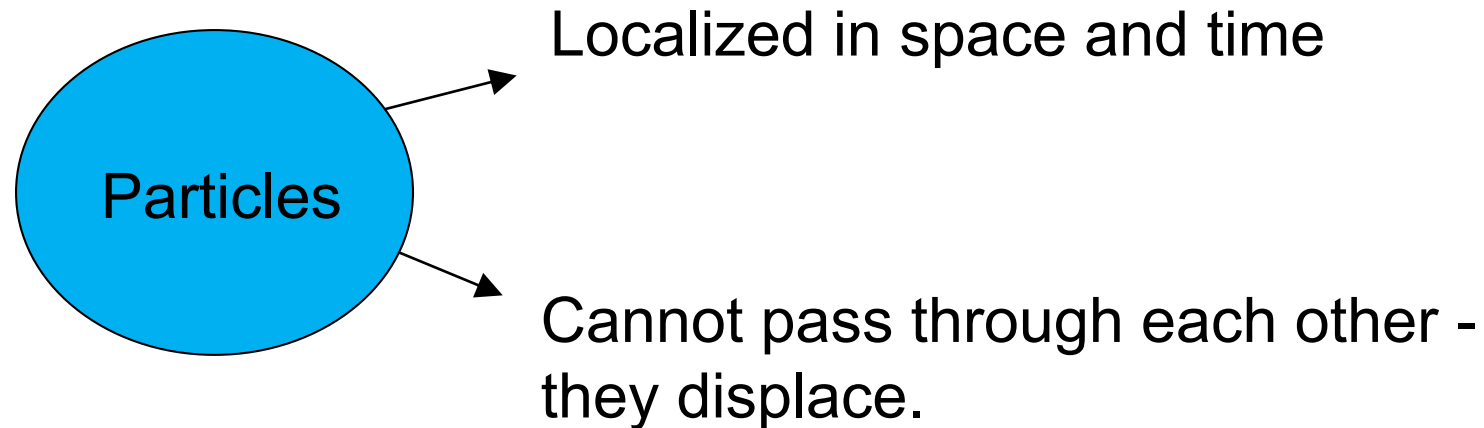
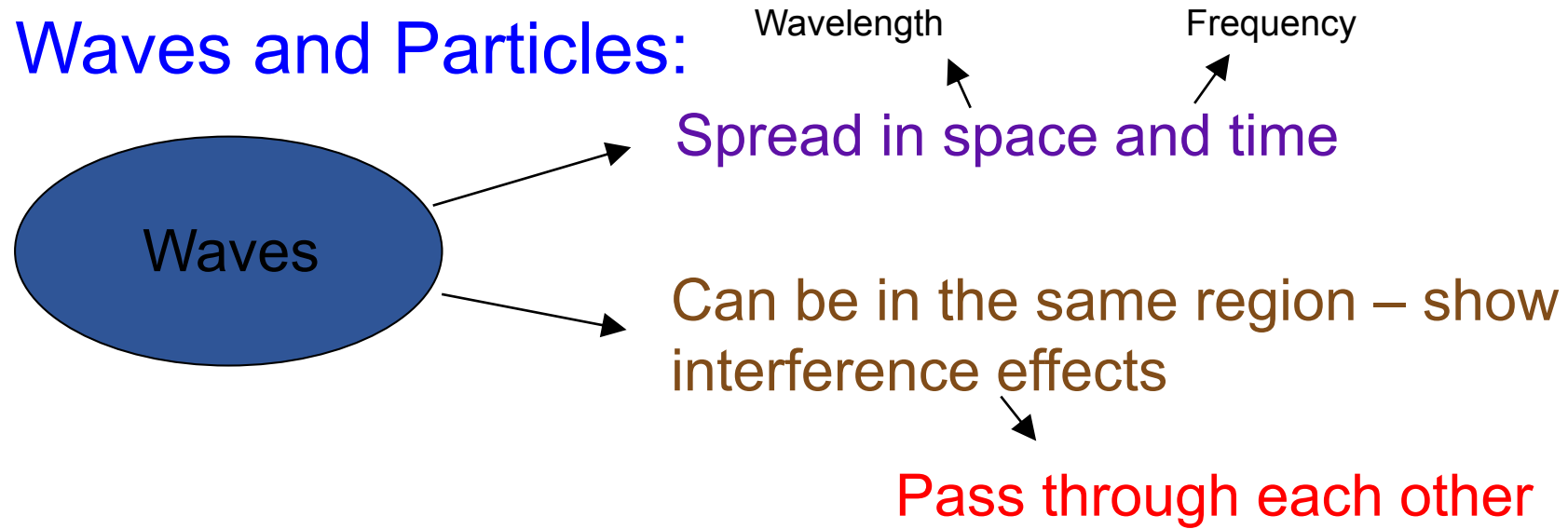


Wave

- Wave can pass through each other.
- They can be in the same region or space.
- When they are in same region of space, they can enhance each other and they can cancel each other. Thus, they can interfere.
- They regain their original form when they pass each other.



Waves and Particles:



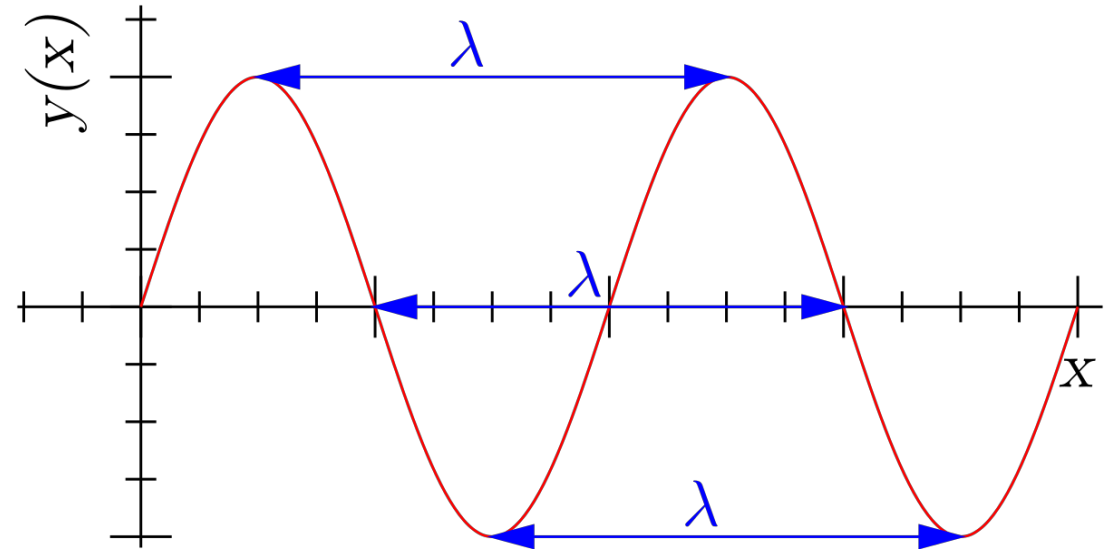
Characteristic Properties of the wave

Water waves, Sound waves

Wavelength

Wavelength:

- It is the distance between adjacent crest.
- It is the distance between adjacent trough.
- The shortest distance between two points that are moving in a phase.



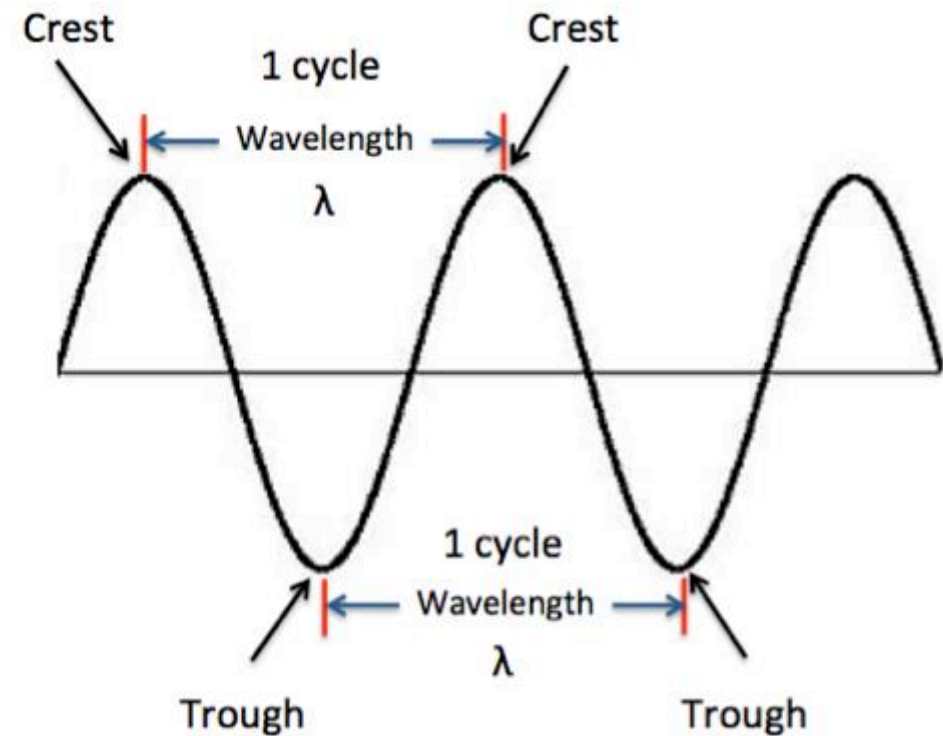
Wavenumber

- It is reciprocal of wavelength, representing the number of wavelengths per unit length.

$$\bar{\nu} = \frac{1}{\lambda}$$

The period of a wave

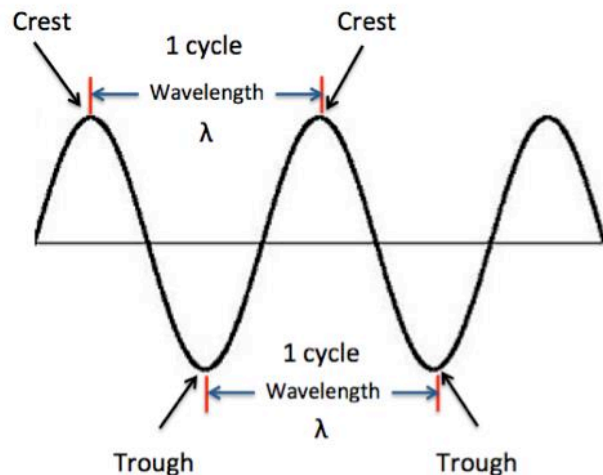
- It is the time taken to complete one full cycle i.e. from crest to crest.
- It is the reciprocal of the frequency.



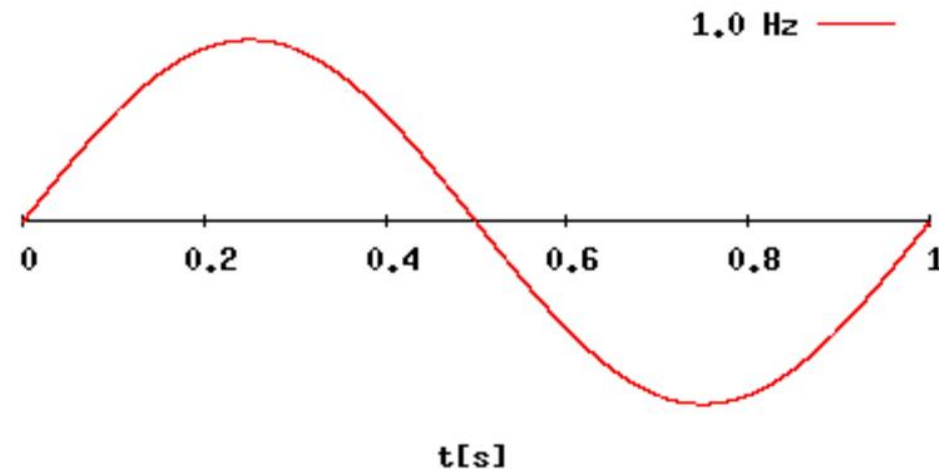
Crest, Trough and Wavelength

The frequency of a wave

- Number of cycles per unit time
- The number of waves that pass crest per unit time
- A wave whose crests passes a fixed point every second has a frequency of 1 Hz.

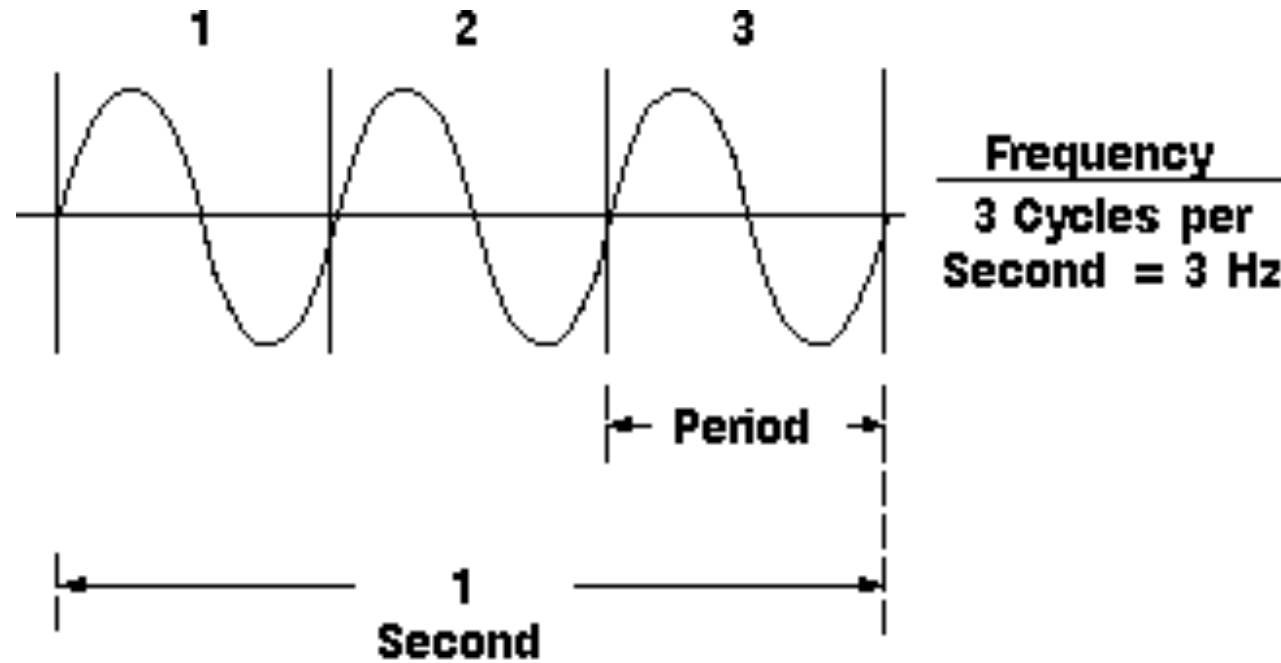


Crest, Trough and Wavelength



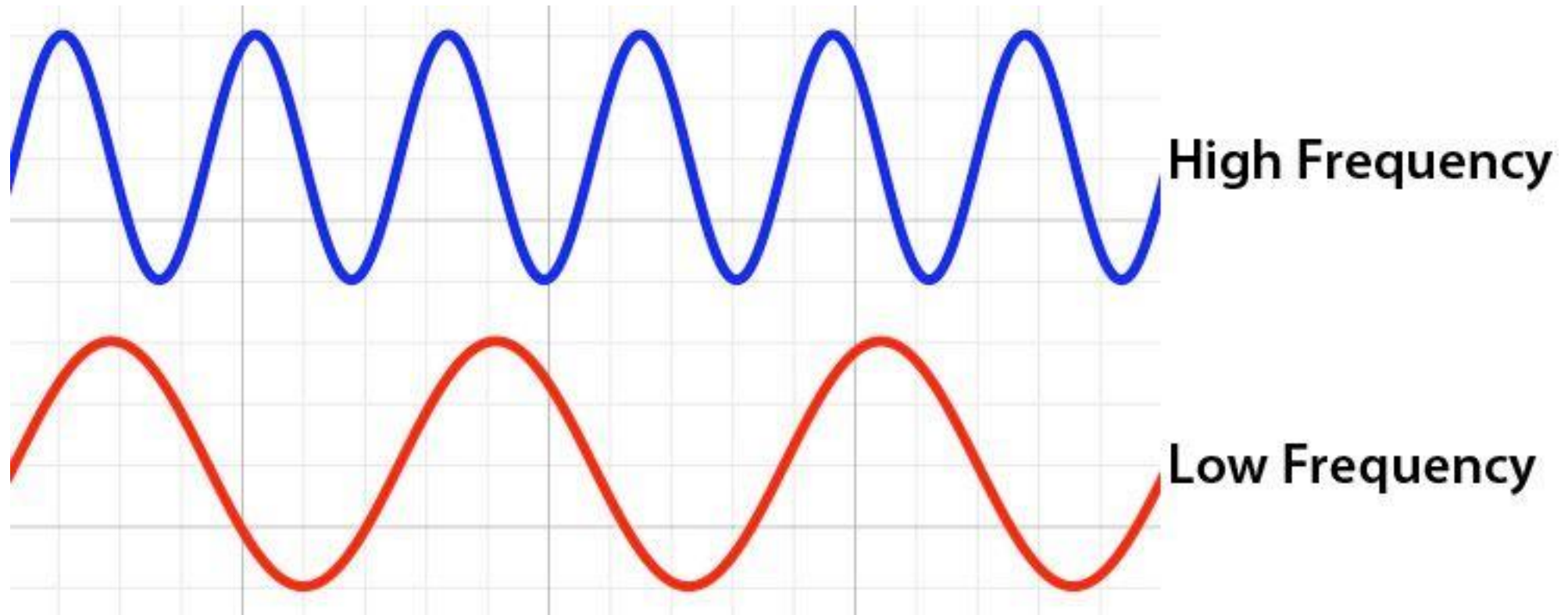
1 Hz equals 1 wave cycle per second

Frequency



https://www.hobbyprojects.com/oscilloscope_tutorial/waveform_measurements.html

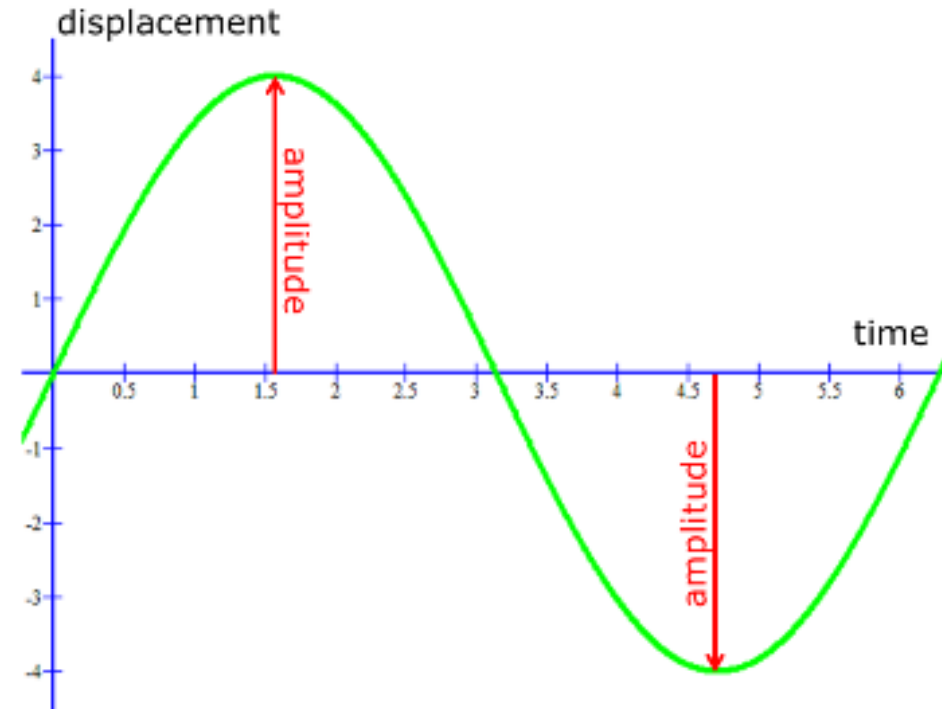
Frequency



<https://www.pinterest.com/pin/325877723012873296/>

Amplitude of the wave

- Amplitude is the height, or maximum displacement, of the wave crest above the undisturbed position.

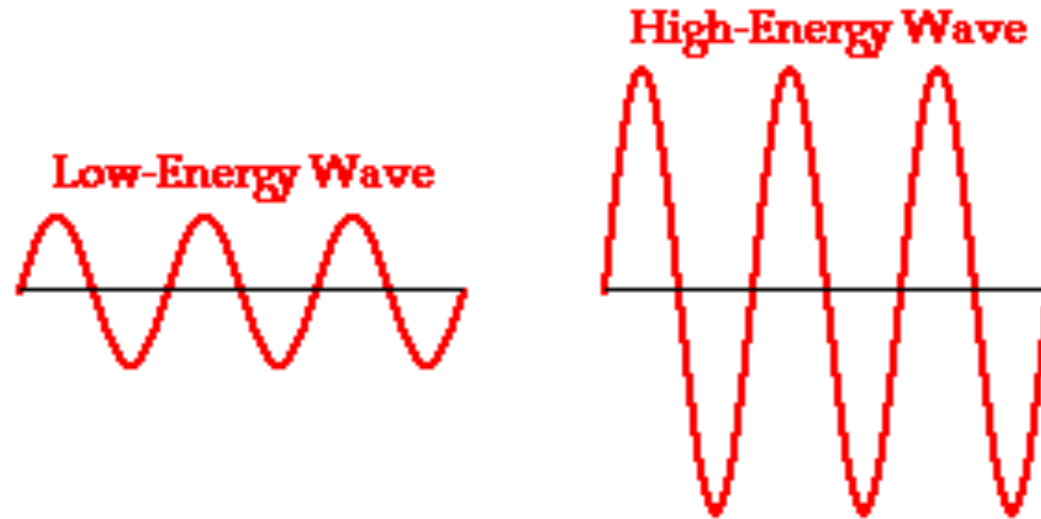


<https://study.com/academy/lesson/the-resultant-amplitude-of-two-superposed-waves.html>

The velocity of the wave

- The velocity of a wave is speed of the wave in a given direction.

Amplitude of the wave



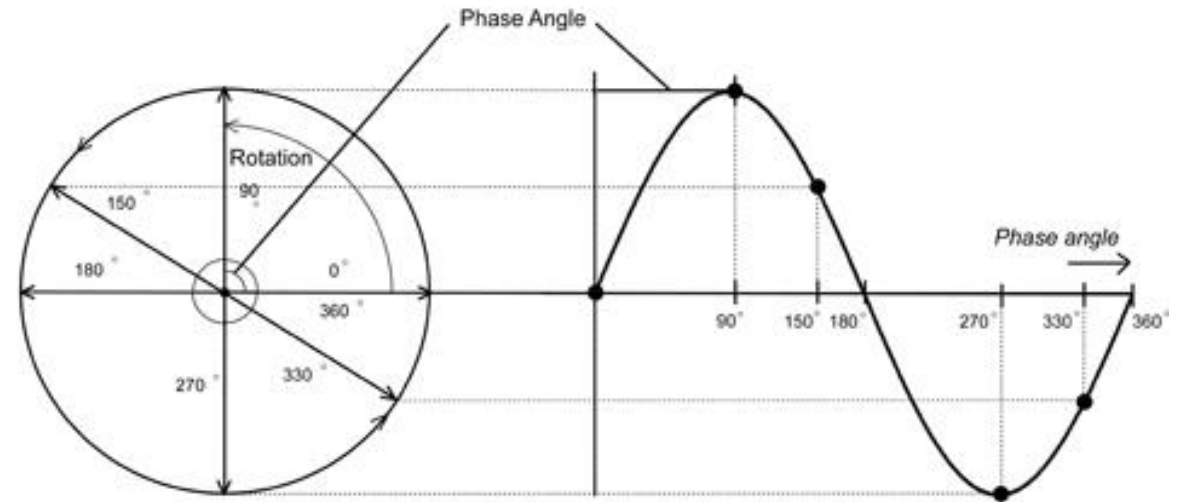
The amplitude of a wave is related to the energy which it transports.

The AMPLITUDE of a wave is related to the energy it transports.

<https://study.com/academy/lesson/the-resultant-amplitude-of-two-superposed-waves.html>

Phase

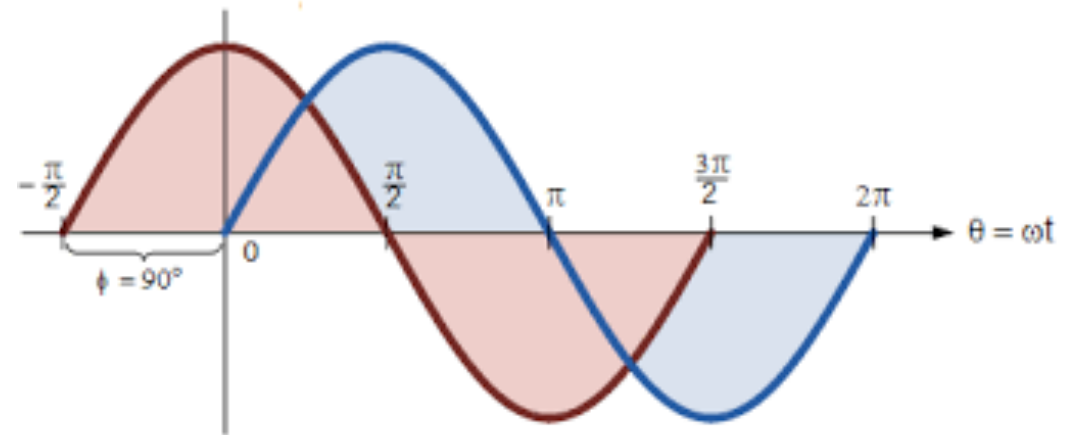
- Two points moving with the same velocity, which have same displacement from the undisturbed resting state, are said to be in phase.
- The phase can be expressed in angle.
- One cycle corresponds to 360 degree.



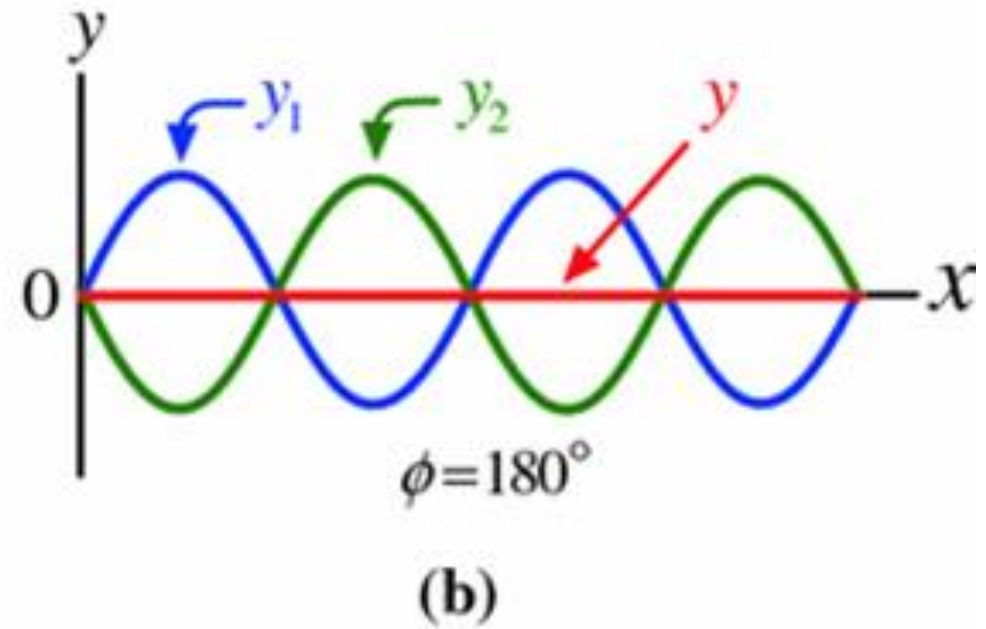
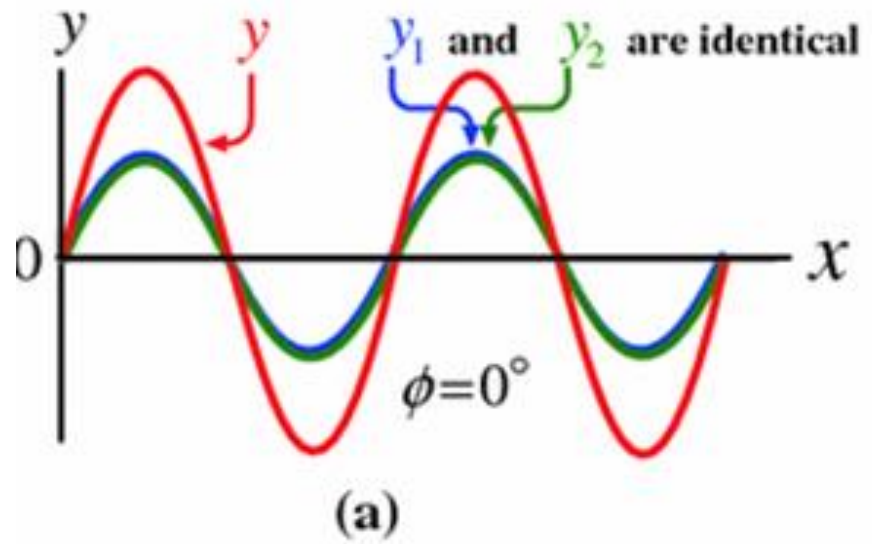
<https://physics.stackexchange.com/questions/142888/wave-displacement-with-phase-difference-180>

Phase difference and Path Difference

- Phase Difference is used to describe the difference in degrees or radians when two or more alternating quantities reach their maximum or zero values.



Phase Difference



Properties shown by both
particle and wave

Particles and Waves Reflected by a Mirror

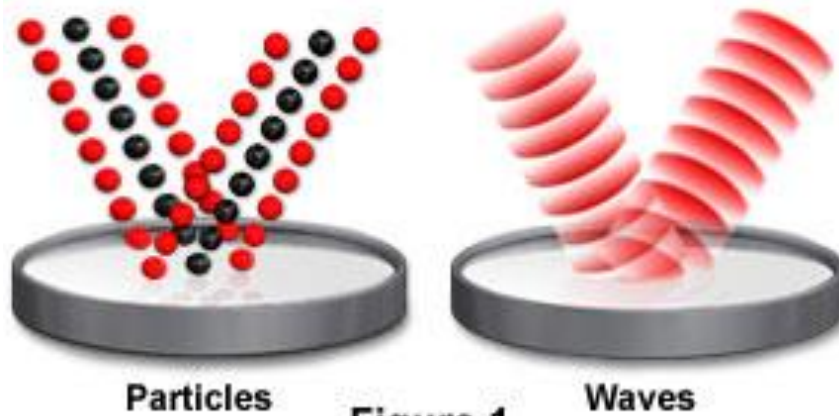


Figure 1

<https://www.olympus-lifescience.com/en/microscope-resource/primer/java/particleorwave/reflection/>

Refraction of Particles and Waves

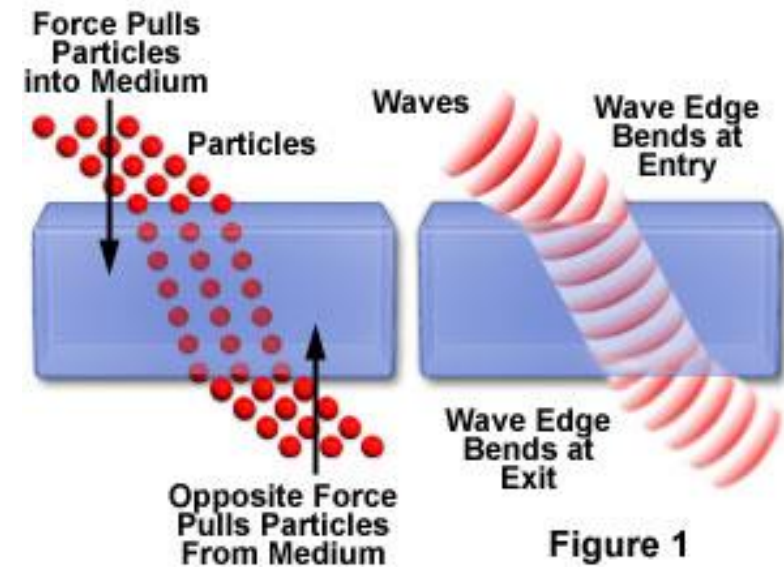


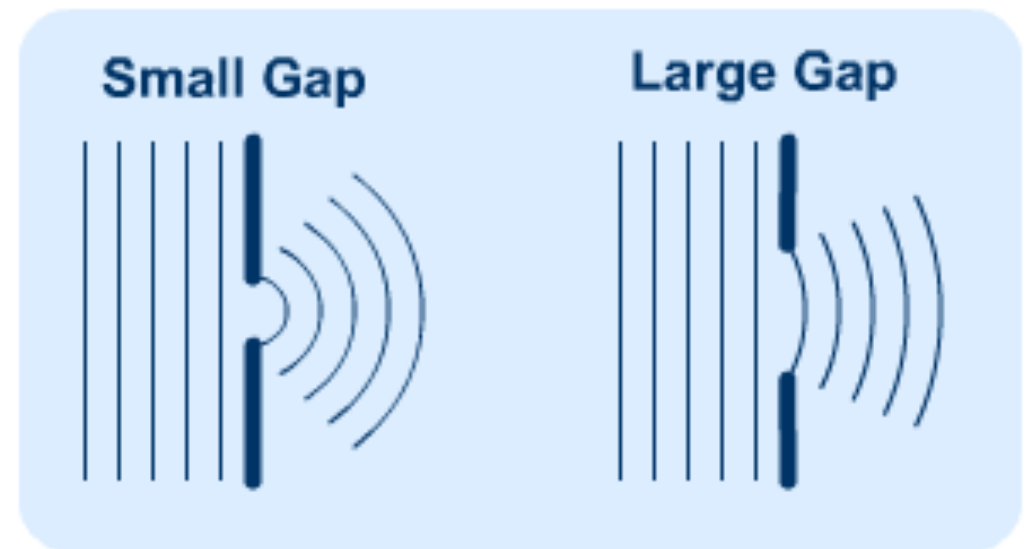
Figure 1

<https://micro.magnet.fsu.edu/primer/java/particleorwave/refraction/index.html>

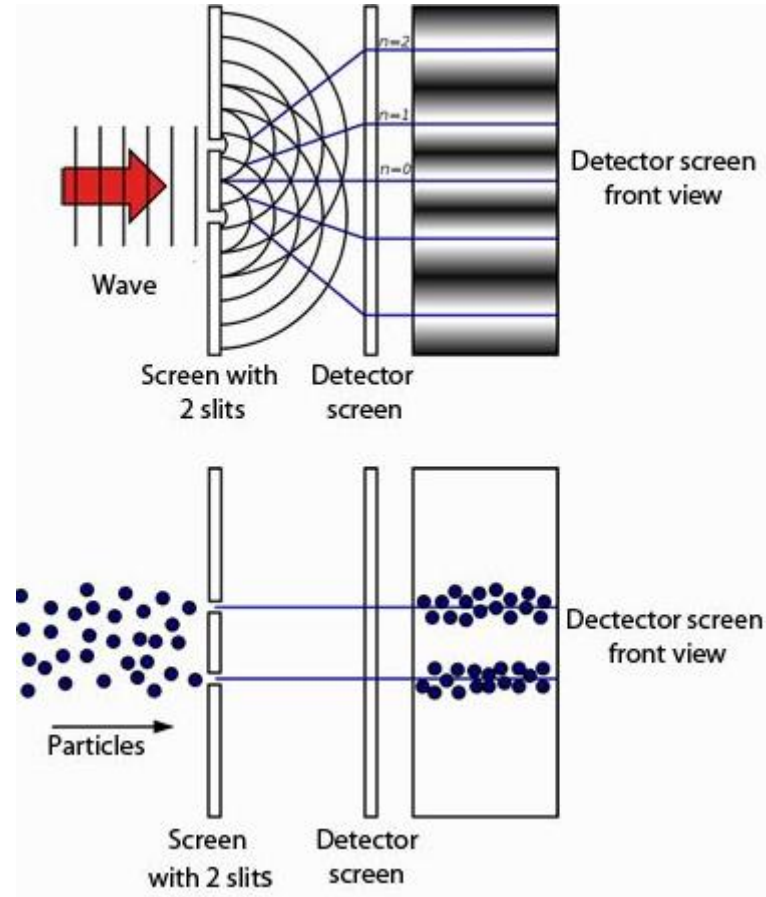
Properties shown by wave
alone

Diffraction

- **As the waves go through the slit, they spread out, this is called diffraction.**
- **Diffraction is most noticeable when the gap size is about the same as the wavelength of the wave.**
- **Light waves have a very short wavelength compared to water waves.**
- **Therefore to diffract light the gap needs to be extremely small, in fact around one thousandth of a millimetre.**

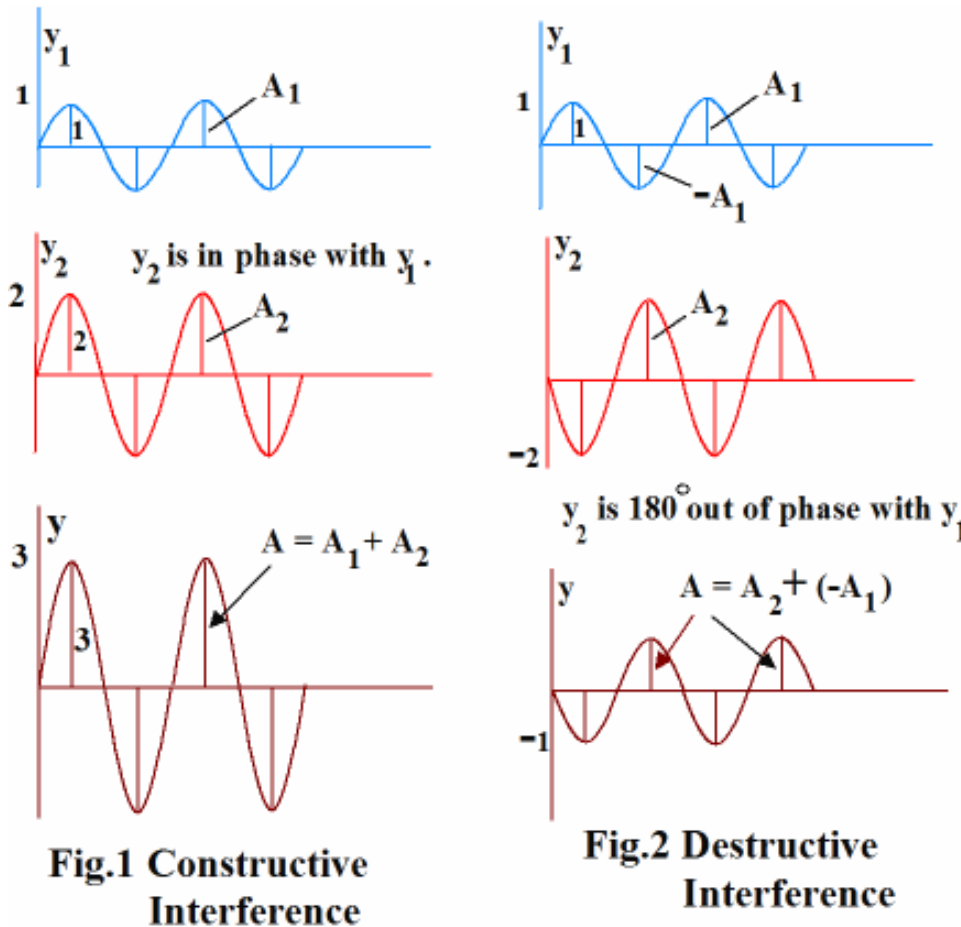


Interference



A diffraction pattern of alternating dark and bright fringes is observed

Constructive and Destructive Interference

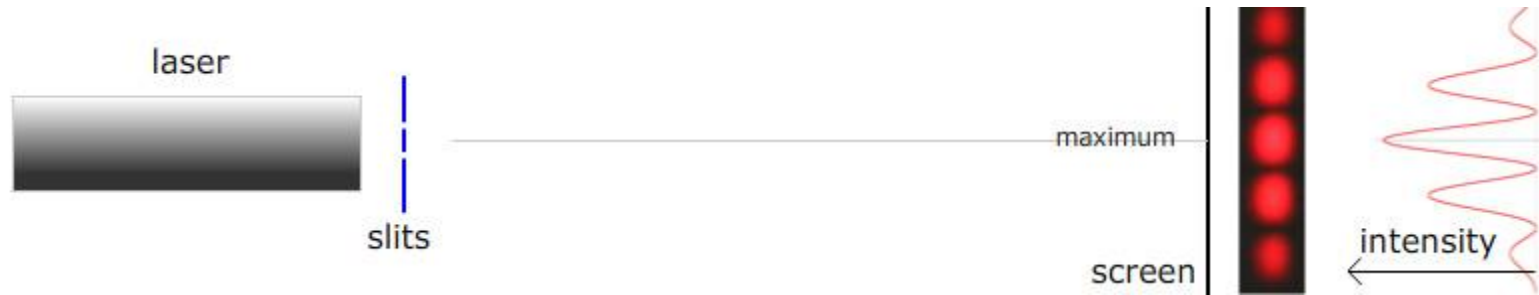


- Constructive Interference: When Crest of one wave interferes with crest of other wave.
- Destructive Interference: When Crest of one wave interferes with trough of other wave.

EMR (wave or particle ??)

Young's Double Slit experiment (1801)

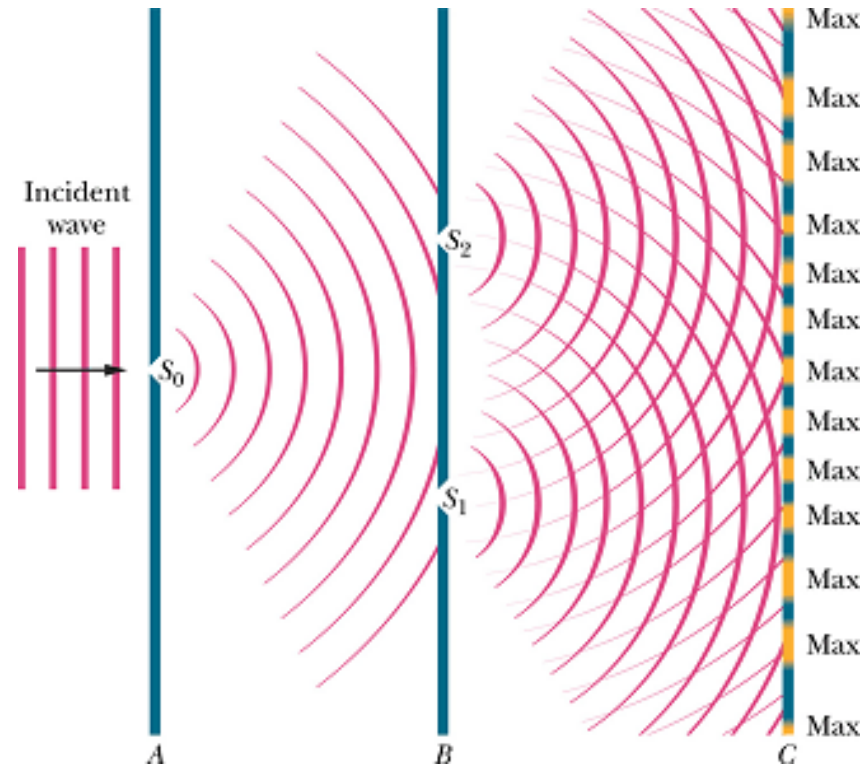
Monochromatic light is allowed to pass through two narrow slits close to each other, and are detected at a screen placed on the other side of the slit.



A diffraction pattern of alternating dark and bright fringes is observed

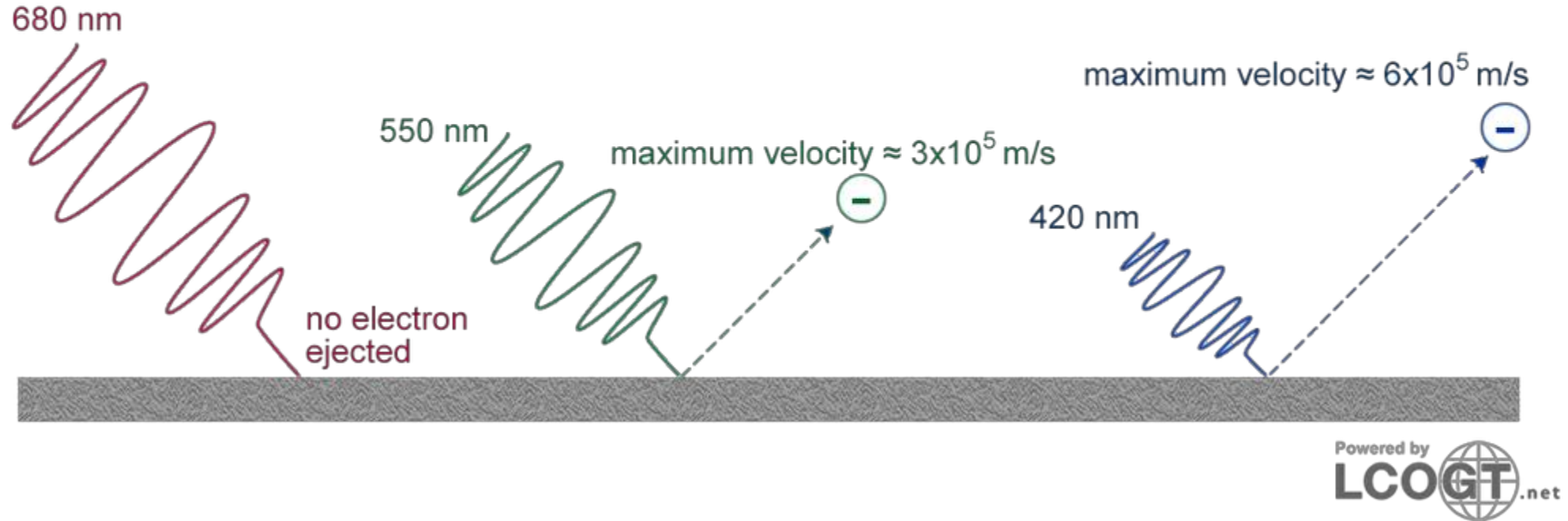
Young Double Slit Experiment

(Wave Nature of Light)



The Photoelectric Effect

Potassium ejects electrons when struck by photons with at least 2.3 eV of energy. This corresponds to a wavelength of about 540 nm, which is visible green light.



RED LIGHT EJECTS SLOWER ELECTRONS THAN BLUE LIGHT

<https://lco.global/spacebook/light/light-particle/>

Photoelectric effect

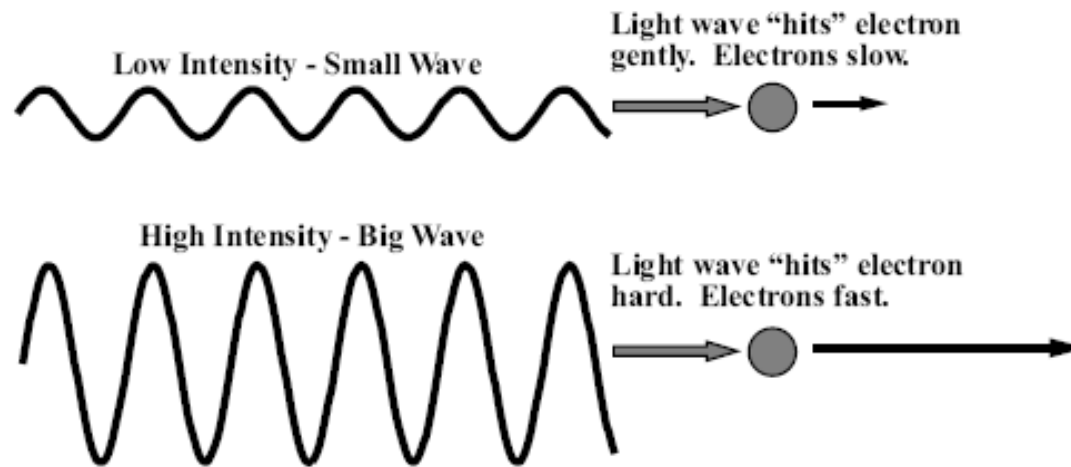
- For a particular metal and a given color of light, say blue, it is found that the electrons come out with a well-defined speed, and that the number of electrons that come out depends on the intensity of the light.
- If the intensity of light is increased, more electrons come out, but each electron has the same speed, independent of the intensity of the light.
- If the color of light is changed to red, the electron speed is slower, and if the color is made redder and redder, the electrons' speed is slower and slower.
- For red enough light, electrons cease to come out of the metal.

Analogy to a man standing in sea water

- A small wave will not knock the person off of his feet
- If the waves get bigger and bigger, eventually they will be big enough to overcome binding of the person's feet from ocean floor.
- The bigger the wave, the farthest the swimmer is thrown out on the beach.



If light behaves as a wave

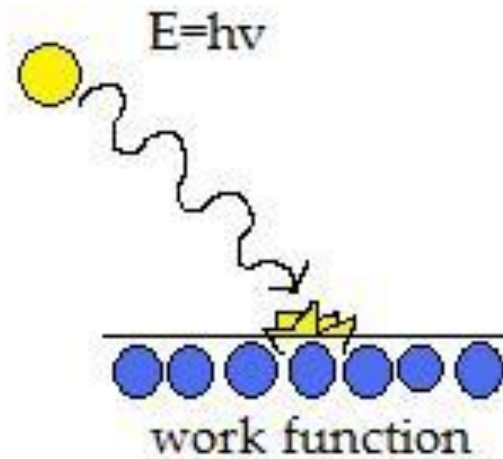


Particle Theory (Albert Einstein, 1905)

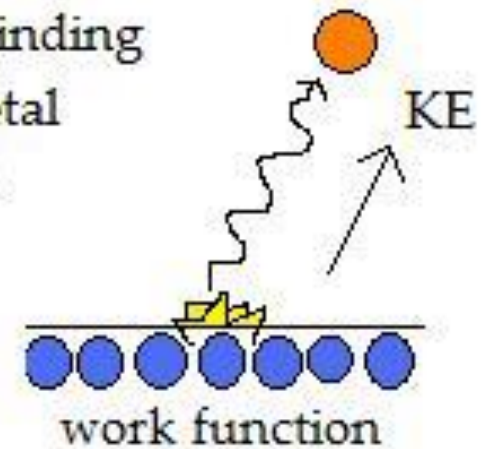
- Electromagnetic radiation or light is composed of many photons, which is a discrete particle.
- One photon hits up one electron and knocks it out of metal.

Einstein (1905)

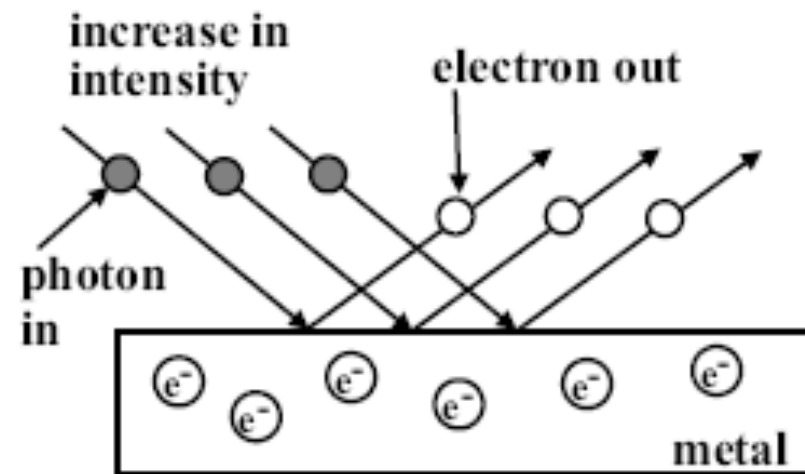
Bound electron
absorbs photon of
energy ($h\nu$)



$h\nu >$ energy binding
electron to metal
surface (work
function) =
photoelectron
released



Effect of Increase in Intensity



Wave-Particle Theory (Albert Einstein)

According to Einstein, a photon resembles a particle but moves like a wave.

The theory suggests that waves of photons traveling through space or matter make up electromagnetic radiation.

The particle and wave properties are not mutually exclusive, they complement each other.

Relation between particle and wave(De Broglie, 1924)

Energy of photon is related to
frequency/wavelength which is a wave attribute.

$$\lambda = \frac{h}{p}$$

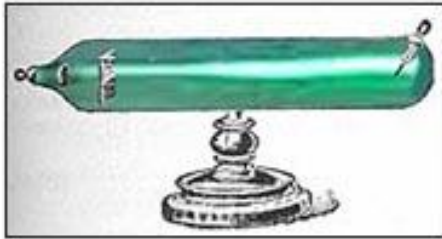
Particles (electrons,
baseball....)

Electrons as a particle



THOMSON

J.J. Thomson Established electron as a fundamental particle of nature. He measured its charge to mass ratio using a Crooke's tube.



Crooke's tube:
Evacuated tube
Visualization of
electron beam.



Animation of electrons moving
and being deflected by an electric
or magnetic field.

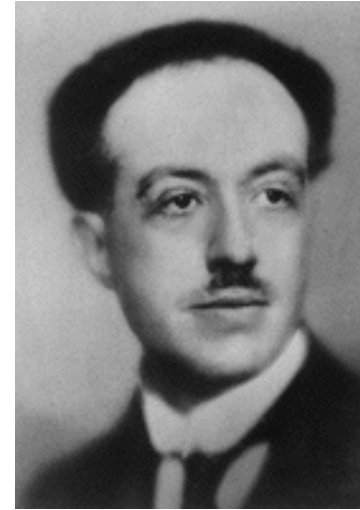
Can material particles exhibit wave nature ?

Can photons, electrons and
baseball exhibits wave nature?

Wave nature of materials

Prince Louis de Broglie while
doing his Ph.D. research
proposed that

Matter particles should exhibit
wave properties just as light
exhibited particle property.

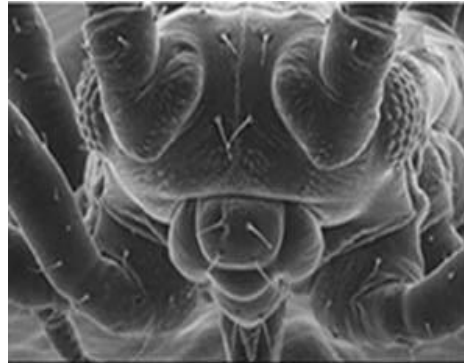


Got Nobel Prize for Physics in 1929

Wave Nature of Matter

These waves were observed a few years later by Davisson and G.P. Thomson with high energy electrons. These electrons show the same pattern when scattered from crystals as X-rays of similar wave lengths.

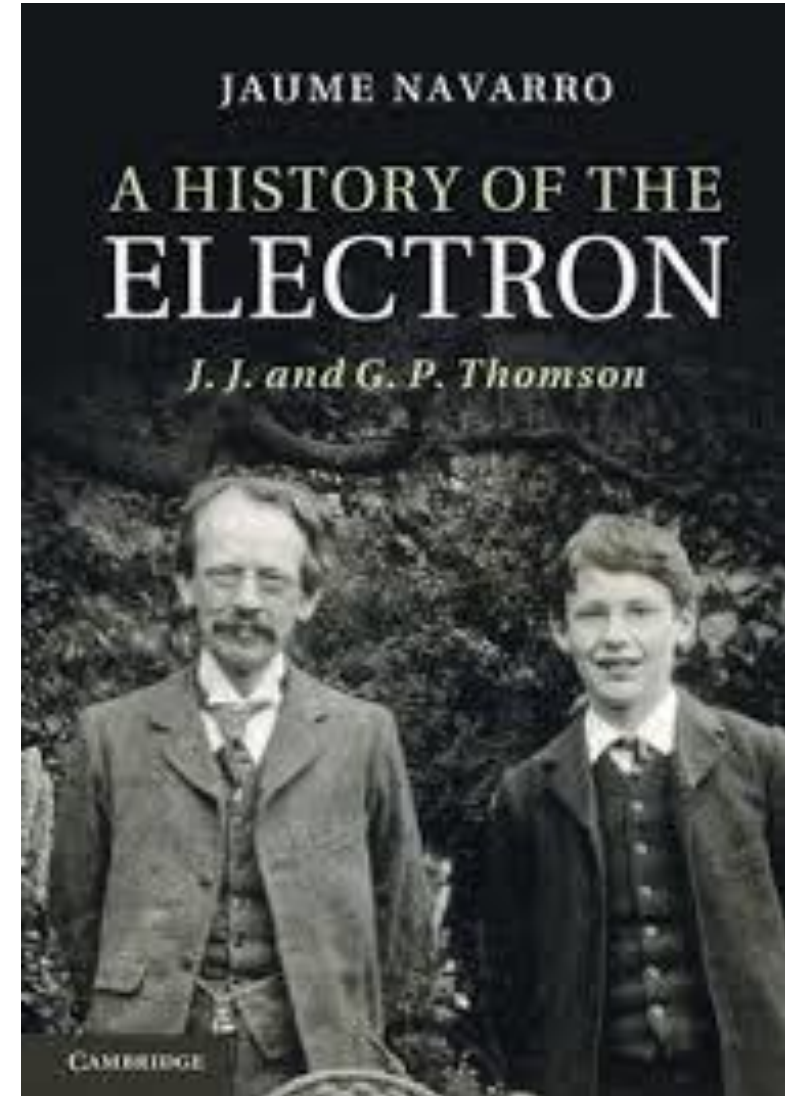
Electron microscope
picture of a fly



<https://www.physics.uci.edu/~silverma/quantum.ppt>

Electron as a wave and particle

- JJ showed that the electron is a particle and GP showed that it is not, or rather that the electron can also behave as a wave.
- JJ got Nobel Prize for Physics in 1906.
- GP got Nobel Prize for Physics in 1927.



Does DeBroglie equation apply to all particle

DeBroglie
Wavelength

$$\lambda = \frac{h}{p}$$

Does this relationship apply to all particles? Consider a pitched baseball:

 $\longrightarrow$ $v = 40 \text{ m/s} = 90 \text{ mi/hr}$
 $m = 0.15 \text{ kg}$ $\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34} \text{ J s}}{(0.15 \text{ kg})(40 \text{ m/s})} = 1.1 \times 10^{-34} \text{ m}$

10^{-10} m
Atomic
diameter
 10^{-14} m
Nuclear
Diameter

For an electron accelerated through 100 Volts: $v = 5.9 \times 10^6 \text{ m/s}$

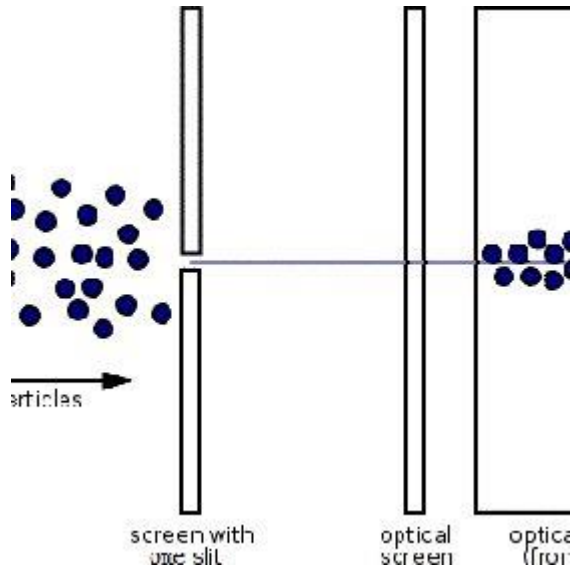
$$\lambda = \frac{6.626 \times 10^{-34} \text{ J s}}{(9.11 \times 10^{-31} \text{ kg})(5.9 \times 10^6 \text{ m/s})} = 1.2 \times 10^{-10} = 0.12 \text{ nm}$$

This is on the order of atomic dimensions and is much shorter than the shortest visible light wavelength of about 390 nm.

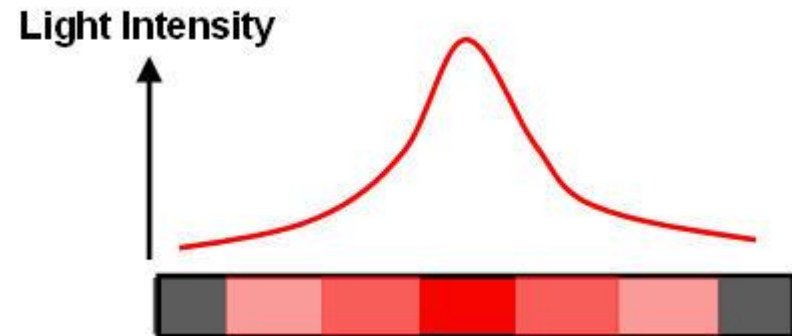
- The de Broglie wavelength λ for macroscopic particles are negligibly small
- This effect is extremely important for light particles, like electrons.

Diffraction of particle

Diffraction of Bullets



Diffraction of Photons



Wave or Particle (Size ??)

- Objects that are large in the absolute sense have the property that the wavelengths associated with them are completely negligible compared to their size. Therefore, large particles only manifest their particle nature, they never manifest their wave nature.
- Electrons and photons are particle and waves, whereas baseball are only particle.

Wavefunction

- Quantum mechanics acknowledges the wave-particle duality of matter by supposing that, rather than traveling along a definite path, a particle is distributed through space like a wave. The wave that in quantum mechanics replaces the classical concept of particle trajectory is called a wavefunction, ψ (“psi”).

A **wave function** in [quantum mechanics](#) describes the [quantum state](#) of an isolated system of one or more particles. There is *one* wave function containing all the information about the entire system, not a separate wave function for each particle in the system.